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Systems and Methods for Fine Adjustment of Roof Models

Abstract

Systems and methods for fine adjustment of computerized roof models are provided. The system generates a **3**D roof structure model based on at least one image obtained from an aerial imagery database. Alternatively, the system could retrieve at least one stored **3**D roof structure model from a **3**D roof structure model database. The system weighs (e.g., scores) each **3**D roof structure model candidate and determines an optimal **3**D roof structure model by applying a variable neighborhood search to a **3**D roof structure model candidate having a highest confidence score among the weighed **3**D roof structure model candidates.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/197,852 filed on May 16, 2023, now U.S. Pat. No. 12,307,593 issued on May 20, 2025, which is a continuation of U.S. application Ser. No. 17/355,845 filed on Jun. 23, 2021, now U.S. Pat. No. 11,651,552 issued on May 16, 2023, which claims priority to U.S. Provisional Patent Application Ser. No. 63/042,813 filed on Jun. 23, 2020, the entire disclosures of which is hereby expressly incorporated by reference.

BACKGROUND
Technical Field
[0002] The present disclosure relates generally to the field of computer modeling of structures. More particularly, the present disclosure relates to systems and methods for fine adjustment of roof models.
Related Art
[0003] Accurate and rapid identification and depiction of objects from digital images (e.g., aerial images, satellite images, etc.) is increasingly important for a variety of applications. For example, information related to various features of buildings, such as roofs, walls, doors, etc., is often used by construction professionals to specify materials and associated costs for both newly-constructed buildings, as well as for replacing and upgrading existing structures. Further, in the insurance industry, accurate information about structures may be used to determine the proper costs for insuring buildings/structures.
[0004] Various software systems have been implemented to process aerial images and/or overlapping image content of an aerial image pair to generate a three-dimensional (3D) model of a building present in the images and/or a 3D model of the structures thereof (e.g., a roof structure). However, these systems have drawbacks, such as missing camera parameter set information associated with each aerial image and an inability to provide a higher resolution estimate of a position of each aerial image (where the aerial images overlap) to provide a smooth transition for display or computation. This may result in an inaccurate 3D roof structure model that consequently does not substantially align with a roof structure present in each aerial image. As such, the ability to refine one or more parameters of a 3D roof structure model to optimize a position and/or orientation of a 3D roof structure model projected onto an aerial image is a powerful tool.
[0005] Thus, what would be desirable is a system that automatically and efficiently refines one or more parameters of a 3D roof structure model to optimize a position and/or orientation of a 3D roof structure model on an aerial image. Accordingly, the systems and methods disclosed herein solve these and other needs.
SUMMARY
[0006] This present disclosure relates to systems and methods for fine adjustment of roof models. The system generates a 3D roof structure model based on at least one image obtained from an aerial imagery database. Alternatively, the system could retrieve at least one stored 3D roof

structure model from a 3D roof structure model database. The system weighs (e.g., scores) each 3D roof structure model candidate and determines an optimal 3D roof structure model by applying a variable neighborhood search to a 3D roof structure model candidate having a highest confidence score among the weighed 3D roof structure model candidates.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The foregoing features of the invention will be apparent from the following Detailed Description of the Invention, taken in connection with the accompanying drawings, in which:

[0008] FIG. **1** is a diagram illustrating hardware and software components capable of being utilized to implement the system of the present disclosure;

[0009] FIGS. **2**A-**2**C are diagrams illustrating a 3D model of a roof structure, a projection of the 3D model onto an input image, and an optimized projection of the 3D model onto the input image;

[0010] FIGS. **3**A-**3**C are graphs illustrating different strategies utilized by a trajectory-based metaheuristic methods;

[0011] FIG. **4** is a flowchart illustrating overall processing steps carried out by the system of the present disclosure;

[0012] FIG. **5** is a flowchart illustrating step **52** of FIG. **4** in greater detail;

[0013] FIGS. **6**A-**6**C are diagrams illustrating the parametric roof representation of a gable roof;

[0014] FIGS. **7**A-**7**C are diagrams illustrating a weighting function carried out by the system of the present disclosure;

[0015] FIG. **8** is a flowchart illustrating processing step **54** of FIG. **4** in greater detail;

[0016] FIG. **9** is a flowchart illustrating processing step **110** of FIG. **8** in greater detail;

[0017] FIGS. **10**A-**10**E are diagrams illustrating the processing steps of FIG. **9**;

[0018] FIG. **11** is a flowchart illustrating step **56** of FIG. **4** in greater detail;

[0019] FIGS. **12**A-**12**D are diagrams illustrating 3D model optimization processing results generated by the system of the present disclosure;

[0020] FIGS. **13**A-**13**D are diagrams illustrating 3D model optimization processing results generated by the system of the present disclosure; and

[0021] FIG. **14** a diagram illustrating another embodiment of the system of the present disclosure.

DETAILED DESCRIPTION

[0022] The present disclosure relates to systems and methods for improved modeling of roof structures, as described in detail below in connection with FIGS. **1-14**. The embodiments described below are related to the optimization of a 3D roof structure model projected onto an image, including, but not limited to, a two-dimensional aerial or satellite image, a three-dimensional image, a three-dimensional point cloud, a three-dimensional light detection and ranging (LiDAR) point cloud, etc.

[0023] By way of background, a 3D roof structure model can be inferred from a plurality of images and/or a plurality of three-dimensional constructs, such as point clouds or LiDAR data. For example, a 3D roof structure model can be constructed based on features of a roof structure present in a plurality of input images. In complex scenarios, the constructed 3D roof structure model can comprise errors such that the 3D roof structure model does not substantially align with the plurality of input images. To mitigate this issue, at least one parameter of the constructed 3D roof structure model can be refined to adjust the 3D roof structure model and improve an alignment thereof with each input image. In particular, a metaheuristic-based approach can be utilized to minimize an adjustment error of the 3D roof structure model during a final step of a 3D model construction process.

[0024] Referring to FIG. **1**, the system **10** of the present disclosure provides for minimizing an adjustment error of a 3D roof structure model. In particular, the system **10** weighs (e.g., scores) a plurality of 3D roof structure model candidates and applies a variable neighborhood search **183** (as shown in FIG. **11**) to a 3D roof structure model candidate having a highest score to optimize the 3D roof structure model candidate. FIG. **1** is a diagram illustrating hardware and software components capable of being utilized to implement the system **10** of the present disclosure. The system **10** could be embodied as a central processing unit **12** (e.g., a hardware processor) coupled to a 3D roof structure model database **14** and/or an aerial imagery database **16**. The system **10** could retrieve at least one stored 3D roof structure model candidate from the 3D roof structure model database **14**. Alternatively, and as discussed below, the system **10** could generate at least one 3D roof structure model candidate based on at least one image obtained from the aerial imagery database **16**. The aerial imagery database **16** could include digital images and/or digital image datasets comprising aerial images, satellite images, etc. Further, the datasets could include, but are not limited to, images of residential and commercial buildings. Even further, the database **16** could store one or more three-dimensional representations of an imaged location (including structures at the location), such as point clouds, LiDAR files, etc., and the system could operate with such three-dimensional representations. As such, by the terms “image” and “imagery” as used herein, it is meant not only optical imagery (including aerial and satellite imagery), but also three-dimensional imagery and computer-generated imagery, including, but not limited to, LiDAR, point clouds, three-dimensional images, etc.

[0025] The hardware processor **12** executes system code which generates and optimizes a 3D roof structure model candidate based on at least one aerial image, point cloud, LiDAR file, etc., received from the aerial imagery database **16**. For example, FIGS. **2**A-C respectively illustrate the generation and optimization of a 3D roof structure model candidate. In particular, FIG. **2**A illustrates a generated 3D roof structure model candidate **32***a* of a building **30** and FIG. **2**B illustrates a 3D roof structure wireframe model **32***b* (corresponding to model **32***a*) projected onto the building **30** present in an input image **34**. It is noted that the wireframe could also be projected onto a three-dimensional representation of the building, such as a point cloud or a LiDAR data. As shown in FIG. **2**B, line segments **36***a*, **36***b*, and **36***c* illustrate that the 3D roof structure wireframe model **32***b* is misaligned with the input image **34**. Accordingly, and as shown in FIG. **2**C, the system **10** generates and optimizes, based on the 3D roof structure wireframe model **32***b*, a 3D roof structure wireframe model **38** that is aligned with the input image **34**. It is noted that the hardware processor could include, but is not limited to, a personal computer, a laptop computer, a tablet computer, a smart telephone, a server, and/or a cloud-based computing platform.

[0026] Referring back to FIG. **1**, the system **10** includes system code **18** (i.e., non-transitory, computer-readable instructions) stored on a computer-readable medium and executable by the hardware processor **12** or one or more computer systems. The code **18** could include various custom-written software modules that carry out the steps/processes discussed herein, and could include, but is not limited to, a 3D roof structure model generator **20***a*, a 3D roof structure model candidate weight function generator **20***b* and an optimal 3D roof structure model module **20***c*. The code **18** could be programmed using any suitable programming languages including, but not limited to, C, C++, C #, Java, Python or any other suitable language. Additionally, the code **18** could be distributed across multiple computer systems in communication with each other over a communications network, and/or stored and executed on a cloud computing platform and remotely accessed by a computer system in communication with the cloud platform. The code **18** could communicate with the 3D roof structure model database **14** and/or aerial imagery database **16**, which could be stored on the same computer system as the code **18**, or on one or more other computer systems in communication with the code **18**.

[0027] Still further, the system **10** could be embodied as a customized hardware component such as a field-programmable gate array (“FPGA”), application-specific integrated circuit (“ASIC”), embedded system, or other customized hardware components without departing from the spirit or scope of the present disclosure. It should be understood that FIG. **1** is only one potential configuration, and the system **10** of the present disclosure can be implemented using a number of different configurations.

[0028] As described in further detail below, the system **10** utilizes a metaheuristic method to optimize a 3D roof structure model candidate. In particular, the system **10** utilizes a variable neighborhood search (VNS) metaheuristic method. A metaheuristic method is a high-level procedure designed to find, generate, or select a heuristic (partial search algorithm) that may provide a solution to an optimization problem, especially with incomplete or imperfect data or limited computational capacity. Example metaheuristic methods include, but are not limited to, Ant Colony Optimization (ACO), Evolutionary Algorithms (EA) which includes Genetic Algorithms (GA) and Memetic Algorithm (MA), Greedy Randomized Adaptive Search (GRASP), Iterated Local Search (ILS), Path Relinking (PR), Simulated Annealing (SA), Scatter Search (SS), Tabu Search (TS) and VNS. VNS is considered a trajectory-based metaheuristic method because the search process follows a trajectory in the solution space. Therefore, given a candidate solution (i.e., an initial solution) a trajectory-based method is able to define a trajectory in the search space using movement operations. The followed trajectory can reveal information about the behavior and effectiveness of the algorithm. FIGS. **3**A-**3**C are graphs illustrating different strategies utilized by trajectory-based metaheuristic methods to avoid local optima when endeavoring to reach a global optimum. In particular, FIG. **3**A illustrates a multi-start search, FIG. **3**B illustrates a random search, and FIG. **3**C illustrates a deterministic search.

[0029] VNS avoids a local optimum by changing a neighborhood structure where a local search procedure is applied. Accordingly, VNS can manage a set of neighborhood structures. Different VNS implementations are known but the VNS metaheuristic algorithm introduced by Pierre Hansen in 2001 comprises the following: [0030] Initialization: select a set of neighborhood structures N.sub.k (k=1, . . . k.sub.max), find an initial solution x, set k=1, choose a stopping condition, [0031] Step 1 (shaking stage): Generate x′∈N.sub.k(x) randomly; [0032] Step 2 (local search): Apply a local search method starting with x′ to find local optimum x″; and [0033] Step 3 (move or not): If x″ is better than the incumbent, then set x=x″ and k=1, otherwise set k=k+1 (or if k=k.sub.max set k=1); and return to Step 1.

[0034] The local search procedure of Step 2 refers to a heuristic algorithm which endeavors to improve a candidate solution. In particular, a local search algorithm commences from a candidate solution and subsequently advances to improved candidate solutions following an iterative process. The process ends when an improved candidate solution cannot be further improved. Different local search procedures are known including, but not limited to, the first improvement local search procedure and the best improvement local search procedure. The first improvement local search procedure selects a candidate solution, explores a neighborhood of the candidate solution and selects a first movement that realizes an improvement over the candidate solution. In contrast, the best improvement local search procedure selects a candidate solution, evaluates all possible solutions in a neighborhood of the candidate solution and selects the best solution among the possible solutions. Each of the first improvement local search procedure and the best improvement local search procedure end when the solution cannot be further improved.

[0035] The application of VNS to transform a local minimum 3D roof structure model into an optimal 3D roof structure model requires a parametric representation of a roof structure to represent it as an n-dimensional variable and a defined weight function which provides for comparing 3D roof structure model candidates to determine a best 3D roof structure model candidate. Generating a parametric representation of a roof structure and defining a weight function will respectively be discussed in further detail below in reference to FIGS. **5** and **6**A-**6**C and FIGS. **8**-**10**E.

[0036] FIG. **4** is a flowchart illustrating overall processing steps **50** carried out by the system **10** of the present disclosure. Beginning in step **52**, the system **10** generates a 3D roof structure model. The system **10** could generate the 3D roof structure model based on at least one image obtained from the aerial imagery database **16**. Alternatively, the system **10** could retrieve at least one stored 3D roof structure model from the 3D roof structure model database **14**. Then, in step **54**, the system weighs each 3D roof structure model candidate. Lastly, in step **56**, the system **10** determines an optimal 3D roof structure model based on a best 3D roof structure model candidate among the weighed 3D roof structure model candidates.

[0037] FIG. **5** is a flowchart illustrating step **52** of FIG. **4** in greater detail. In particular, FIG. **5** illustrates processing steps carried out by the system **10** for parameterizing a roof structure present in an aerial image using roof components. Beginning in step **60**, the system **10** receives a user input. The user input can include an identification of one or more roof components present in an aerial image, roof parameters, roof constraints, or any combination thereof. The roof parameters and roof constraints can be used to generate a roof component(s). The input can be received through a command-line interface, a graphical user interface, or any other suitable method. In step **62**, the system **10** inputs the roof components into a geometry creation algorithm ("GCA"). In step **64**, the system generates a constrained 3D geometry via the GCA. Lastly, in step **66**, the system **10** displays a 3D roof structure model candidate. The 3D roof structure model can be composed of points, vertices, line segments, surfaces, etc.

[0038] A parametric roof representation refers to the construction of a roof structure model with different degrees of complexity utilizing a small set of parameters. This representation provides for a constructed 3D roof structure model candidate to adhere to a particular set of rules and geometrical restrictions required for VNS to apply a local search procedure to determine an optimal 3D roof structure model. FIGS. **6**A-**6**C are diagrams illustrating an example parametric roof representation of a gable roof. FIG. **6**A illustrates a generated 3D gable roof structure model **72** of a building **70** and FIG. **6**B illustrates a 2D diagram of the 3D gable roof structure model **72** of FIG. **6**A. As shown in FIG. **6**B, the gable roof comprises two rectangular planes having a plurality of rakes **78**, two eaves **80** and a ridge **82**. The rakes **78** are defined by line segments A-AB, AB-B, D-CD and CD-C, the eaves **80** are defined by line segments A-D and B-C and the ridge **82** is defined by the line segment AB-CD. FIG. **6**C illustrates a parametric representation of the 3D gable roof structure model **72** of FIG. **6**A. As shown in FIG. **6**C, a gable roof is defined by utilizing seven continuous parameters: a ridge vertex location (X.sub.AB, Y.sub.AB, Z.sub.AB), a roof orientation or azimuth (a), a length of the ridge (**1**), a roof width (w) and an eave height (Z.sub.EAVE). Modifying any of these parameters provides for constructing a new 3D gable roof structure model derived from the original 3D gable roof structure model. The same applies to other roof structure types including, but not limited to, a hip-roof, a gablet, a mansard, etc. and to their combinations in arbitrarily complex roofs composed of multiple roof structure types. Accordingly, a complex roof can be described by a fixed-size vector of continuous variables indicative of the parameters controlling and defining a roof structure.

[0039] The system **10** utilizes the VNS metaheuristic method to optimize a 3D roof structure model candidate and, as such, each 3D roof structure model candidate must be scored to determine whether a particular 3D roof structure model candidate can be considered "better than" a previously computed 3D roof structure model candidate. Accordingly, the VNS metaheuristic model requires determining a weight or cost function to evaluate an accuracy of each 3D roof structure model candidate. The system **10** utilizes a weight function that relies on basic image information. Intuitively, a 3D roof structure model candidate projected onto an image should closely resemble the image gradients and roof edges of the roof structure present in the image. The closer the projected roof lines (i.e., 2D segments) of the 3D roof structure model candidate are to those image features, the greater weight will be assigned to the candidate solution.

[0040] FIGS. **7**A-**7**C are diagrams illustrating a weight function carried out by the system **10** of the present disclosure. In particular, FIG. **7**A illustrates a generated 3D roof structure model **82** of a building **80** and FIGS. **7**B and **7**C illustrate carrying out the weight function for the 3D roof structure model **82** for different images views of the building **80**. Referring to FIG. **7**B, image **84***a* illustrates a perspective view of a roof structure **86** of the building **80** and image **84***b* illustrates 2D roof segments **88** corresponding to the roof structure **86** and extracted from the image **84***a*. Lastly, image **84***c* illustrates 3D roof segments **90** corresponding to the 3D roof structure model **82** projected onto the 2D roof segments **88**. Referring to FIG. **7**C, image **94***a* illustrates another perspective view of the roof structure **86** of the building **80** and image **94***b* illustrates 2D roof segments **96** corresponding to the roof structure **86** and extracted from the image **94***a*. Lastly, image **94***c* illustrates 3D roof segments **98** corresponding to the 3D roof structure model **82** projected onto the 2D roof segments **96**. As discussed above, an optimal 3D roof structure model should align with the 2D roof segments extracted from an image.

[0041] FIG. **8** is a flowchart illustrating processing step **54** of FIG. **4** in greater detail, in which the system **10** scores a 3D roof structure model candidate. Beginning in step **100**, the system **10** selects an image having a roof structure present therein. In step **102**, the system **10** extracts 2D roof segments corresponding to the roof structure present in the image. It is noted that traditional computer vision algorithms or complex neural networks can be utilized to extract the 2D roof segments corresponding to the roof structure present in the image.

[0042] In step **104**, the system **10** obtains a generated 3D roof structure model candidate corresponding to the roof structure present in the image or obtains a stored 3D roof structure model candidate from the 3D roof structure model database **14**. Then, in step **106**, the system **10** extracts 3D roof segments from the 3D roof structure model candidate. In particular, given a set of roof parameters, the system **10** transforms the 3D roof structure model candidate into a set of vertices, segments and faces (e.g., a 3D roof structure wireframe model). A segment-based representation is selected for scoring the 3D roof structure model candidate. Next, in step **108**, the extracted 3D candidate roof segments are projected onto the image.

[0043] In step **110**, the system **10** compares the extracted 2D segments from the image and 2D segments of the extracted candidate 3D roof segments projected onto the image. In particular, the system **10** determines a distance between the extracted 2D segments from the image and the extracted 2D segments of the extracted candidate 3D roof segments projected onto the image to determine a score of the 3D roof structure model candidate. Then, the system **10** updates a global score with the determined score of the 3D roof structure model candidate. In step **112**, the system **10** determines whether all images have been processed. If all images have not been processed, then the process returns to step **100** to select and process a new image. Alternatively, if all images have been processed, then the process proceeds to step **114**. In step **114**, the scores of the 3D roof structure model candidates are averaged to yield a weight for the 3D roof structure model candidate. As discussed above, a 3D roof structure model candidate projected onto an image should closely resemble the image gradients and roof edges of the roof structure present in the image. In particular, the smaller the distance between those image features and the extracted 2D segments of the extracted candidate 3D roof segments projected onto the image, the greater weight (i.e., confidence) will be assigned to the 3D roof structure model candidate.

[0044] FIG. **9** is a flowchart illustrating processing step **110** of FIG. **8** in greater detail, in which the system **10** determines distance values between a set of 2D segments extracted from an image and 2D segments of extracted candidate 3D roof segments projected onto the image. Beginning in step **130**, the system **10** subsamples a candidate 2D segment to generate a set of segment points. In step **132**, the system **10** determines a geometric distance to a closest parallel image 2D segment for each segment point. Then, in step **134**, the system **10** sums and averages the determined geometric distances for the segment points to yield a geometric distance value (i.e., a score) for the candidate 2D segment. In step **136**, the system **10** determines whether all candidate 2D segments have been processed. If all candidate 2D segments have not been processed, then the process returns to step **130** to select and process another candidate 2D segment. Alternatively, if all candidate 2D segments have been processed, then the process proceeds to step **138**. In step **138**, the system sums and averages the geometric distance values of the respective candidate 2D segments to yield a score for the 3D roof structure model candidate.

[0045] FIGS. **10**A-**10**E are diagrams illustrating the processing steps of FIG. **9**. FIG. **10**A illustrates a 3D roof structure model candidate **152** of a building **150** and a candidate 2D segment **154** of the 3D roof structure model candidate **152**. FIG. **10**B illustrates a selected image **155** showing a perspective view of a roof structure **156** of the building **150**. FIG. **10**C illustrates extracted 2D roof segments **158** corresponding to the roof structure **156** of the building **150** present in the image **155** of FIG. **10**B. It is noted that traditional computer vision algorithms or complex neural networks can be utilized to extract 2D roof segments from the image **155**.

[0046] FIG. **10**D illustrates a view **161** of a projection of the candidate 2D segment **154** onto the extracted 2D roof segments **158** corresponding to the roof structure **156** of the building **150** present in the image **155** of FIG. **10**B. In particular, FIG. **10**D illustrates a comparison of the 2D roof segment **160** of the extracted 2D roof segments **158** with the projected candidate 2D segment **154**. As shown in FIG. **10**D, the projected candidate 2D segment **154** is parallel to but not aligned with the 2D roof segment **160**. FIG. **10**E illustrates a magnified view of the view **161** of FIG. **10**D. As shown in FIG. **10**E, the system **10** subsamples the candidate 2D segment **154** to generate a set of segment points **164** and determines a geometric distance **166** to the 2D roof segment **160** for each segment point **164**.

[0047] FIG. **11** is a flowchart illustrating step **56** of FIG. **4** in greater detail, in which the system **10** applies VNS **183** using a best improvement local search procedure to transform a 3D roof structure model candidate (i.e., a sub-optimal candidate) to an optimal 3D roof structure model. It is noted that VNS **183** can utilize other search procedures including, but not limited to, first improvement, neighborhood change, etc. Beginning in step **180**, the system **10** obtains a generated 3D roof structure model candidate corresponding to a roof structure present in a set of images **180** or obtains a stored 3D roof structure model candidate from the 3D roof structure model database **14**. The 3D roof structure model candidate may not align with the roof structure present in each image (i.e., the 3D roof structure model candidate is a sub-optimal solution).

[0048] In step **184**, given the set of images **180** and the 3D roof structure model candidate **182** as inputs, the system **10** applies VNS **183** using the best improvement local search procedure. In particular, the neighborhood structure is initialized to N.sub.1 (i.e., k=1) and all possible solutions in the neighborhood structure N**1** are weighted. It is noted that for VNS, the neighborhood structure refers to a number of parameters to be modified simultaneously. Therefore, when VNS **183** executes the best improvement local search procedure over the 3D roof structure model candidate utilizing the neighborhood structure N.sub.1, only one roof parameter is modified to yield a 3D roof structure model candidate solution. Accordingly, a neighborhood structure N.sub.2 requires the modification of two roof parameters, a neighborhood structure N.sub.3 requires the modification of three roof parameters, a neighborhood structure N.sub.4 requires the modification of four roof parameters, and so on. A parameter modification refers to the application of an offset to a current parameter value. It is noted that each parameter can have a different offset value based on a respective unit of measurement. For example, a ridge_length parameter expressed in meters should be increased or decreased using a different offset value than an azimuth parameter expressed in radians.

[0049] In step **186**, VNS **183** determines whether a candidate solution in the neighborhood structure N.sub.1 improves upon the 3D roof structure model candidate (i.e., the incumbent candidate). If VNS **183** determines that a candidate solution in the neighborhood structure N.sub.1 improves upon the 3D roof structure model candidate, then VNS **183** replaces the 3D roof structure model candidate with the neighborhood structure N.sub.1 candidate solution. Then, the process returns to step **184** to execute a new best improvement local search procedure over the neighborhood structure N.sub.1 candidate solution utilizing the neighborhood structure N.sub.1. Alternatively, if the system **10** determines that a candidate solution in the neighborhood structure N.sub.1 does not improve upon the 3D roof structure model candidate, then the process proceeds to

step **188**.
[0050] In step **188**, VNS **183** determines whether a maximum number of neighborhood structures (i.e., k=k.sub.max) has been utilized. If the maximum number of neighborhood structures has not been utilized, then VNS **183** increments the neighborhood structure (i.e., k=k+1). Subsequently, the process returns to step **184** to execute a new best improvement local search procedure over the neighborhood structure N.sub.1 candidate solution utilizing a neighborhood structure N.sub.2. It is understood that if VNS **183** determines that a candidate solution in the neighborhood structure N.sub.2 improves upon the neighborhood structure N.sub.1 candidate solution, then the process returns to the initial neighborhood structure N.sub.1 to execute a new best improvement local search procedure over the neighborhood structure N.sub.2 candidate solution. Alternatively, if the maximum number of neighborhood structures has been utilized, then the process proceeds to step **190**. In step **190**, an optimal 3D roof structure model is realized. In particular, the optimal 3D roof structure model is indicative of an iteration that cannot be further improved and an improvement of the 3D roof structure model candidate.
[0051] FIGS. **12**A-**12**D and FIGS. **13**A-**13**D are diagrams illustrating 3D roof structure model optimization processing results generated by the system **10** of the present disclosure, in which VNS **183** has been applied to a plurality of 3D roof structure model candidates **212**, **222**, **232**, **242**, **252**, **262**, **272** and **282**. In particular, FIGS. **12**A-**12**D respectively illustrate 3D roof structure model candidates **212**, **222**, **232** and **242** corresponding to a roof structure of building **211** in respective image views **210***a*, **220***a*, **230***a* and **240***a* and their corresponding optimal 3D roof structure models **216**, **226**, **236** and **246** in respective image views **210***b*, **220***b*, **230***b* and **240***b*.
[0052] For example, FIG. **12**A illustrates that line segment **214** of 3D roof structure model candidate **212** does not substantially align with the roof structure of building **211** in image **210***a* but that after application of VNS **183** to the 3D roof structure model candidate **212**, the generated optimal 3D roof structure model **216** aligns with the roof structure of building **211** in image **210***b*. FIG. **12**B illustrates that line segment **224** of 3D roof structure model candidate **222** does not substantially align with the roof structure of building **211** in image **220***a* but that after application of VNS **183** to the 3D roof structure model candidate **222**, the generated optimal 3D roof structure model **226** aligns with the roof structure of building **211** in image **220***b*. FIG. **12**C illustrates that line segment **234** of 3D roof structure model candidate **232** does not substantially align with the roof structure of building **211** in image **230***a* but that after application of VNS **183** to the 3D roof structure model candidate **232**, the generated optimal 3D roof structure model **236** aligns with the roof structure of building **211** in image **230***b*. FIG. **12**D illustrates that line segment **244** of 3D roof structure model candidate **242** does not substantially align with the roof structure of building **211** in image **240***a* but that after application of VNS **183** to the 3D roof structure model candidate **242**, the generated optimal 3D roof structure model **246** aligns with the roof structure of building **211** in image **240***b*.
[0053] FIGS. **13**A-**13**D respectively illustrate 3D roof structure model candidates **252**, **262**, **272** and **282** corresponding to a roof structure of building **251** in respective image views **250***a*, **260***a*, **270***a* and **280***a* and their corresponding optimal 3D roof structure models **256**, **266**, **276** and **286** in respective image views **250***b*, **260***b*, **270***b* and **280***b*. For example, FIG. **13**A illustrates that line segment **254** of 3D roof structure model candidate **252** does not substantially align with the roof structure of building **251** in image **250***a* but that after application of VNS **183** to the 3D roof structure model candidate **252**, the generated optimal 3D roof structure model **256** aligns with the roof structure of building **251** in image **250***b*. FIG. **13**B illustrates that line segment **264** of 3D roof structure model candidate **262** does not substantially align with the roof structure of building **251** in image **260***a* but that after application of VNS **183** to the 3D roof structure model candidate **262**, the generated optimal 3D roof structure model **266** aligns with the roof structure of building **251** in image **260***b*. FIG. **13**C illustrates that line segment **274** of 3D roof structure model candidate **272** does not substantially align with the roof structure of building **251** in image **270***a* but that after

application of VNS **183** to the 3D roof structure model candidate **272**, the generated optimal 3D roof structure model **276** aligns with the roof structure of building **251** in image **270***b*. FIG. **13**D illustrates that line segment **284** of 3D roof structure model candidate **282** does not substantially align with the roof structure of building **251** in image **280***a* but that after application of VNS **183** to the 3D roof structure model candidate **282**, the generated optimal 3D roof structure model **286** aligns with the roof structure of building **251** in image **280***b*.

[0054] FIG. **14** a diagram illustrating another embodiment of the system **300** of the present disclosure. In particular, FIG. **14** illustrates additional computer hardware and network components on which the system **300** could be implemented. The system **300** can include a plurality of internal servers **302***a*-**302***n* having at least one processor and memory for executing the computer instructions and methods described above (which could be embodied as system code **18**). The system **300** can also include a plurality of image storage servers **304***a*-**304***n* for receiving image data and/or video data. The system **300** can also include a plurality of camera devices **306***a*-**306***n* for capturing image data and/or video data. For example, the camera devices can include, but are not limited to, a unmanned aerial vehicle **306***a*, an airplane **306***b*, and a satellite **306***n*. The internal servers **302***a*-**302***n*, the image storage servers **304***a*-**304***n*, and the camera devices **306***a*-**306***n* can communicate over a communication network **308**. Of course, the system **300** need not be implemented on multiple devices, and indeed, the system **300** could be implemented on a single computer system (e.g., a personal computer, server, mobile computer, smart phone, etc.) without departing from the spirit or scope of the present disclosure.

[0055] Having thus described the system and method in detail, it is to be understood that the foregoing description is not intended to limit the spirit or scope thereof. It will be understood that the embodiments of the present disclosure described herein are merely exemplary and that a person skilled in the art can make any variations and modification without departing from the spirit and scope of the disclosure. All such variations and modifications, including those discussed above, are intended to be included within the scope of the disclosure. What is desired to be protected by Letters Patent is set forth in the following claims.

Claims

1. A method for fine adjustment of a computerized roof model, comprising the steps of: generating by a processor a computerized three-dimensional (3D) roof model; weighing a plurality of candidate 3D roof structure models; selecting one of the plurality of candidate 3D roof structure models having a highest weight among the plurality of candidate 3D roof structure models; optimizing the selected one of the plurality of candidate 3D roof structure models to create an optimal 3D roof structuremodel; and displaying the optimal 3D roof structure model superimposed over an image of a structure.

2. The method of claim 1, wherein the step of optimizing the selected one of the plurality of candidate 3D roof structure models comprises performing a best improvement local search on a set of image lines and at least one of the plurality of candidate 3D roof structure models.

3. The method of claim 1, wherein the step of generating the 3D roof model comprises receiving user input indicating at least one of a roof component present in an aerial image, a roof parameter, or a roof constraint.

4. The method of claim 3, wherein the step of generating the 3D roof model comprises processing roof parameters using a geometry creation algorithm.

5. The method of claim 4, wherein the step of generating the 3D roof model comprises generating a constrained 3D geometry via the geometry creation algorithm.

6. The method of claim 5, wherein the step of generating the 3D roof model comprises displaying a 3D roof structure model candidate.

7. A system for fine adjustment of a computerized roof model, comprising: a database storing a plurality of candidate 3D roof structure models; and a processor in communication with the database, the processor programmed to perform the steps of: generating a computerized three-dimensional (3D) roof model; weighing the plurality of candidate 3D roof structure models; selecting one of the plurality of candidate 3D roof structure models having a highest weight among the plurality of candidate 3D roof structure models; optimizing the selected one of the plurality of candidate 3D roof structure models to create an optimal 3D roof structure model; and displaying the optimal 3D roof structure model superimposed over an image of a structure.

8. The system of claim 7, wherein the step of optimizing the selected one of the plurality of candidate 3D roof structure models comprises performing a best improvement local search on a set of image lines and at least one of the plurality of candidate 3D roof structure models.

9. The system of claim 7, wherein the step of generating the 3D roof model comprises receiving user input indicating at least one of a roof component present in an aerial image, a roof parameter, or a roof constraint.

10. The system of claim 9, wherein the step of generating the 3D roof model comprises processing roof parameters using a geometry creation algorithm.

11. The system of claim 10, wherein the step of generating the 3D roof model comprises generating a constrained 3D geometry via the geometry creation algorithm.

12. The system of claim 11, wherein the step of generating the 3D roof model comprises displaying a 3D roof structure model candidate.